

Count-Only Inverse-Variance Weighting for Intervention Harvesting

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Abstract

Intervention-harvesting estimators of position bias [1] combine per-cell click-rate estimates across many $(q, d, \text{position})$ cells whose impression counts span several orders of magnitude. Under the Position-Based Model (PBM), with cell-level relevance treated as homoscedastic, the *harmonic* weight $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ is the count-only inverse-variance choice—Aitken’s classical generalised-least-squares solution [2] restricted to count-only weights—and an unconditional Cauchy–Schwarz inequality shows it minimises the delta-method variance *surrogate* $V_k + V_{k'}$ for any impression counts. We propose the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ of the self-normalised weighted ratio estimator [11, 13] as a finite-sample diagnostic that is well aligned with the conditional-on-counts efficiency question, in contrast to a naive pair-bootstrap. On the full KDD Cup 2012 Track 2 sponsored-search log (235M impressions, 4.6M interventional pairs), the plug-in asymptotic relative efficiency of harmonic over uniform is 4.53, well-calibrated against click-resample standard error (1–2% at two sample fractions), and robust to weight-capping (ARE ≥ 4.1 at the 99.99th percentile). A Cochran- Q diagnostic confirms substantial PBM mis-specification ($Q/\text{df} = 3.83$), allowing efficiency comparisons to be interpreted separately from model fit; under such mis-specification different weight schemes target different weighted averages of cell-specific ratios—an estimand caveat we discuss explicitly. Synthetic (43%–54% MSE reduction at high imbalance) and Yahoo-LTR semi-synthetic ($\sim 40\%$ variance reduction at $r = 50$) experiments with ground truth corroborate the efficiency prediction.

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1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [8]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1, 2)$), **60.7% of pairs**

have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions [9]. Public benchmarks hide this shape: the released Baidu-ULTR [16] aggregation has 94% of (q, d, pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$ *because* the publication pipeline deduplicates impressions per cell—an artefact of release engineering, not of the underlying production logs. The unaggregated KDD release preserves the heavy-tailed production distribution.

Equal-weighting heteroscedastic estimators is statistically sub-optimal by a century of classical theory: Aitken’s generalised-least-squares theorem [2] mandates inverse-variance weights, a result the meta-analysis literature [5, 7] has championed for fifty years. Recent ULTR work [1, 14] introduced *harmonic* weighting $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ to reduce variance versus uniform on heavy-tailed counts, but existing validation has relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated heavy-tailed data (Section 3).

Contributions. (1) We identify harmonic as the count-only inverse-variance choice under the standard PBM with homoscedastic relevance (Theorem 2), prove an unconditional Cauchy–Schwarz reduction of the delta-method variance *surrogate* $V_k + V_{k'}$ (Theorem 3, conditional on the surrogate objective and unconditional in the counts), and characterise the exact asymmetric surrogate oracle of which harmonic is the $\alpha = \beta$ special case (Theorem 4). (2) We adopt the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ as a finite-sample efficiency diagnostic aligned with the conditional-on-counts question, and use empirical click-resample SE as its calibration baseline; we discuss why pair-bootstrap is misaligned with this question on heavy-tailed data. (3) On the full KDD log we measure $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, calibrated within 1–2% across two sample fractions, and robust to weight-capping at the 99.99th percentile ($\text{ARE} \geq 4.1$). We quantify PBM mis-specification via Cochran’s $Q/\text{df} = 3.83$ and discuss the estimand caveat that follows. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Under PBM, $P(\text{click} \mid q, d, k) = p_k r_{q,d}$. Given an interventional set $I_{k,k'}$ of (q, d) pairs that appeared at both positions k and k' , the position-bias ratio is estimated as [1]

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}_k^i}{\sum_i \omega_i \hat{\theta}_{k'}^i}, \quad \hat{\theta}_k^i = \frac{C_k^i}{N_k^i}. \quad (1)$$

This is a self-normalised weighted ratio estimator. We say a weight scheme is *ratio-unbiased* when the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$, i.e. $\mathbb{E}[A \mid N]/\mathbb{E}[B \mid N] = p_k/p_{k'}$ (not $\mathbb{E}[\hat{R}] = p_k/p_{k'}$; the finite-sample bias of the self-normalised ratio is $O(1/N)$ under standard regularity). Under PBM, the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$ for any count-only weight $\omega_i = \omega(N_k^i, N_{k'}^i)$, and

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the resulting self-normalised ratio estimator is consistent for this target [1].

PROPOSITION 1 (CONDITIONAL DELTA-METHOD VARIANCE). *By the delta method [13], $\text{Var}(\hat{R} \mid \mathbf{N}) \approx R^2[\alpha V_k(\omega) + \beta V_{k'}(\omega)]$ with $V_r(\omega) = \sum_i \omega_i^2 / N_r^i / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, $\beta = (1 - p_{k'})/p_{k'}$.*

Harmonic as count-only IVW. For Bernoulli observations, $\text{Var}(\hat{\theta}_k^i) = \theta_k^i(1 - \theta_k^i)/N_k^i$. Under PBM, $\theta_k^i = p_k r_i$, so the Bernoulli variance contains a cell-level relevance r_i that is generally heterogeneous. After suppressing this cell-level relevance heterogeneity, the inverse-variance optimum of Aitken [2] restricted to count-only weights is the harmonic mean of the two cell counts. We make this precise as a statement about the variance *surrogate* obtained by ignoring cell-level relevance variation:

THEOREM 2 (COUNT-ONLY IVW ON THE SURROGATE). *Among all non-negative weights $\omega_i = \omega(N_k^i, N_{k'}^i)$, harmonic weighting minimises the count-only delta-method variance surrogate $V_k(\omega) + V_{k'}(\omega)$.*

THEOREM 3 (SURROGATE GUARANTEE, UNCONDITIONAL IN COUNTS). *For any impression counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(1) + V_{k'}(1)$. The guarantee is unconditional in the counts but conditional on $V_k + V_{k'}$ as the objective; it implies improved $\text{Var}(\hat{R} \mid \mathbf{N})$ under PBM when $\alpha \approx \beta$ (Proposition 1 below).*

SKETCH. Substituting $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2) \sum_i 1/\omega_i^h$. Cauchy-Schwarz gives $n^2 \leq W \sum_i 1/\omega_i^h$. $\square$

THEOREM 4 (EXACT ASYMMETRIC ORACLE). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ with $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, attained at $\omega_i^* \propto N_k^i N_{k'}^i / (\alpha N_{k'}^i + \beta N_k^i)$. Harmonic is the $\alpha = \beta$ special case.

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count heuristic of [1], which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into ω^* yields variance within $\pm 2\%$ of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

Limits of the symmetric guarantee. The unconditional bound in Theorem 3 concerns $V_k + V_{k'}$. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(1) + \beta V_{k'}(1)$ does *not* hold unconditionally: a closed-form $n = 2$ counterexample with $(N_k^1, N_{k'}^1) = (1, 100)$, $(N_k^2, N_{k'}^2) = (100, 1)$ and $(\alpha, \beta) = (0.01, 1)$ has harmonic strictly worse than uniform on the asymmetric surrogate. The configurations under which this fails are degenerate (each cell informative at exactly one position); we have not encountered them in any realistic count distribution we tested. The ω^* in Theorem 4 remains the exact oracle for all (α, β) .

3 Plug-in Asymptotic Variance: the Right Instrument

$\tilde{V}$ in closed form. With $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at k and k' conditionally independent given counts, the delta method gives

$$\tilde{V}(\omega) = \frac{\sum_i \omega_i^2 \theta_k^i (1 - \theta_k^i) / N_k^i}{A^2} + \frac{\sum_i \omega_i^2 \theta_{k'}^i (1 - \theta_{k'}^i) / N_{k'}^i}{B^2}, \quad (2)$$

Table 1: Plug-in SE vs. click-resample SE across KDD subsamples, and ARE(harmonic, uniform) at each scale. Click resample only computed at small f for runtime; agreement is within 1–2% where measured. ARE grows with data volume because heavy-tail mass becomes more pronounced at larger scales.

f	pairs	$\widehat{\text{SE}}_U^{\text{plug}}$	$\widehat{\text{SE}}_U^{\text{boot}}$	$\widehat{\text{SE}}_H^{\text{plug}}$	ARE
10^{-3}	4.3K	0.131	0.132	0.111	1.66
10^{-2}	54K	0.039	0.038	0.028	2.21
10^{-1}	542K	0.012	–	0.007	3.07
1.0	4.61M	0.0042	–	0.0020	4.53

$\text{SE}(\hat{R}) = |\hat{R}| \sqrt{\tilde{V}(\omega)}$. Plugging in $\hat{\theta}_i$ (with light shrinkage toward the marginal) yields the textbook self-normalised IS variance estimator [10–12].

ARE. For schemes a, b , $\text{ARE}(a, b) = \tilde{V}(\omega_b) / \tilde{V}(\omega_a)$: scheme b needs $\text{ARE}(a, b)$ times the data of a for equal precision. $\tilde{V}$ scales as $1/\text{ESS}(\omega)$, with Kong’s effective sample size [10] $\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

Why pair-bootstrap is misaligned here. A naive bootstrap that resamples (q, d) pairs estimates a different quantity than the conditional on-counts efficiency we care about, for three reasons. (i) Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. (ii) Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. (iii) Pair-bootstrap entangles estimator efficiency with model fit; $\tilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration and stability at scale. Table 1 reports plug-in SE versus empirical click-resample SE across KDD sample fractions, together with ARE(harmonic, uniform) at each scale. Plug-in agrees with empirical click-resample SE within 1–2% where the latter is computed, and the ARE grows with sample size ($1.66 \rightarrow 4.53$), because the heavy-tailed structure of the count distribution becomes more pronounced as more data is included rather than averaged out.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \text{ad}, \text{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair (1, 2) interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$.

Headline (Table 2). $\text{ARE}(\text{harmonic, uniform}) = 4.53$. **Uniform requires $\sim 4.5\times$ the impressions of harmonic for matching precision**, with $\text{SE}_U / \text{SE}_H \approx 2.13$. The min heuristic [1] achieves 4.41, the count-only oracle 4.57; harmonic is the principled parameter-free optimum.

Table 2: Plug-in delta-method asymptotic variance $\tilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61\text{M}$). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5 . The full- θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min [1]	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	4.53	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ -aware oracle	2.343	0.148	0.002	1.1

Table 3: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.

$N_1 + N_2$ tertile	n pairs	$\tilde{V}_U$	ARE(h, u)
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	4.98

Stratified ARE (Table 3). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by $5\times$ on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

Weight-cap robustness. The harmonic ESS of 1,849 out of 4.61M pairs invites a concern that the efficiency gain rests on a few extreme cells, but the small ESS reflects deliberate concentration on the pairs containing nearly all statistical information in this heavy-tailed setting rather than instability of the estimator. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS 7.6×10^5); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

Estimand under PBM mis-specification. PBM is a known approximation; on real logs different weight schemes estimate different functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight ω is

$$R_\omega = \frac{\sum_i \omega_i \theta_k^i}{\sum_i \omega_i \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$ (uniform $\rightarrow 1.68$, harmonic $\rightarrow 1.71$, θ -aware oracle $\rightarrow 2.34$ on KDD). The plug-in $\tilde{V}$ comparison is a precision statement at fixed ω —a model-based efficiency comparison—and does *not* claim that one R_ω is the “correct” position-bias ratio. Choosing ω thus involves a precision/estimand trade-off; harmonic’s attractive property is that, among count-only schemes, it concentrates

on high-information cells (where ρ_i is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

Quantifying PBM mis-specification (Cochran’s Q). PBM is a known approximation, not a literal model – prior work [6, 14, 15] has long established that real logs violate it. The relevant question is how large the deviation is. Cochran’s Q on the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2) gives $Q = 6.17\times 10^4$, $\text{df} = 16,131$, $Q/\text{df} = 3.83$: per-cell odds-ratio variability is $\sim 3.8\times$ what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\tilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω .

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1]: $M=10$ positions, $p_k = k^{-\eta}$ with $\eta = 1$, AdjacentChain estimator. A fraction s of documents are “anchors” with heavy traffic at one position ($N \sim \text{Unif}(50r, 200r)$) and light traffic at the other ($N \sim \text{Unif}(1, 51)$); the remaining $1-s$ are balanced ($N \sim \text{Unif}(25, 75)$ at both positions); $s \in \{0.5, \dots, 0.95\}$, $r \in \{1, \dots, 100\}$, 500 trials per cell. At $s = 0.80$, $r = 100$ harmonic reduces variance by 43.6% vs. uniform and 5% vs. min; reduction grows to $\sim 54\%$ at $s = 0.90$, $r = 50$. Across the full (s, r) grid the data-equivalence factor $\text{DEF}(s, r) = b_{\text{uni}}(s, r)/b_{\text{harm}}(s, r)$ – the ratio of base impressions per document at which each scheme first hits a fixed MSE threshold – has median 1.31, mean 1.30, max 1.94. Highly imbalanced cells achieve up to $2\times$ data savings; near-balanced cells ($r = 1$) saturate at $\text{DEF} \approx 1$.

Bias comparability under PBM violations. We additionally simulate three explicit PBM violation models – trust bias, cascade, and position-dependent relevance [1?] – at imbalance ratio 50:1, split 80/20, 5,000 trials. Across all three models the count-only schemes (uniform, min, harmonic) exhibit *comparable* bias², so harmonic’s lower variance translates directly into lower MSE (Table 4). Under cascade, where history-dependent examination introduces a substantial structural mismatch (bias² ≈ 0.011 for every ratio-unbiased scheme), the variance edge is preserved but the practical MSE gain shrinks because bias dominates.

Yahoo-LTR semi-synthetic. Same simulator with graded relevance from the Yahoo Learning to Rank Challenge Set 1 test split (6,936 queries; relevance grades $g \in \{0, \dots, 4\}$ mapped to $r(q, d) = (2^g - 1)/15$ following NDCG; click model $P(C = 1 \mid q, d, k) = p_k r(q, d)$; 200 trials). Harmonic reduces variance by $\sim 40\%$ vs. uniform at imbalance $r = 50$ (Var $1.5\times 10^{-4} \rightarrow 8.7\times 10^{-5}$). Aggressive clipping trades variance for bias and yields $\sim 458\times$ higher MSE, confirming that variance-reduction without bias control is not viable.

Production-style traffic simulation. $K=5$ concurrent rankers; a default ranker shows each document at its natural position, treatment rankers swap adjacent pairs with probability 0.5. Under uniform traffic ($5\times 20\%$) only 0.4% of adjacent pairs exceed impression ratio 10 and harmonic’s MSE reduction is 6.3%; under production traffic ($80\%+4\times 5\%$), 16.7% of pairs exceed ratio 10 and harmonic’s MSE

Table 4: MSE decomposition under three explicit PBM violations (anchor imbalance 50:1, split 80/20, 5,000 trials). Count-only schemes share comparable bias² within each model; harmonic achieves the lowest variance, hence the lowest MSE.

Model	Scheme	MSE	Variance	Bias ²
Trust	Uniform	0.00487	0.00486	1.0e-5
	Min	0.00298	0.00297	2.0e-5
	Harmonic	0.00272	0.00272	0.0
Cascade	Uniform	0.01376	0.00276	0.01100
	Min	0.01242	0.00171	0.01071
	Harmonic	0.01252	0.00147	0.01105
Pos.-dep.	Uniform	0.04086	0.03973	0.00113
	Min	0.03896	0.03770	0.00127
	Harmonic	0.03724	0.03613	0.00111

reduction grows to **17.2%**, confirming the gain scales directly with traffic-induced imbalance.

6 Related Work

Intervention harvesting was introduced by [1] with the uniform and min-count variants we compare against; [14] jointly estimates relevance and propensities via EM. Self-normalised weighted ratio estimators have a long history in importance sampling and survey statistics: Owen [11] §9.2 derives the asymptotic variance we use; Kong [10] introduces ESS; Hesterberg [4] analyses heterogeneous-weight variance. Inverse-variance weighting traces to Aitken [2] and is canonical in meta-analysis [5, 7]. The closest analogue in counterfactual learning is the self-normalised IPS estimator SNIPS [12], which applies the same $\tilde{V}$ framework to off-policy evaluation in bandits; [3] analyses why weight heterogeneity, not sample count, governs effective precision on logged feedback. PBM violations in real logs are documented in [14, 15] and reviewed in [6].

7 Conclusion

Framing harmonic weighting as the count-only inverse-variance optimum (Theorems 2–4) and replacing bootstrap with the plug-in delta-method variance $\tilde{V}$ (Eq. 2) yields rigorous, calibration-verified efficiency comparisons on heavy-tailed real click logs. On the full KDD Cup 2012 Track 2 log, $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, rising to $\sim 5\times$ on information-bearing cells, with PBM misspecification quantified separately by Cochran- Q ($Q/\text{df} = 3.83$). Future work: shrinkage variants of the θ -aware oracle (Theorem 4 with $\hat{\theta}_i$ regularised); click models beyond PBM [6, 14, 15] where the intervention-harvesting family must be reformulated.

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review.

Full code, scripts, and data-processing pipelines are provided as supplementary material.

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